Paper Replication Report #184: Scattered TNO Binary (208996) 2003 AZ84 Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2012, 2019), Benecchi et al. (2011), and Thirouin et al. (2014) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of 2:5 mean-motion resonant / scattered Trans-Neptunian binary (208996) 2003 AZ84 ($R_{\text{primary}} = 360 \pm 20$ km, $R_{\text{secondary}} = 36 \pm 5$ km). Astrometric mutual orbital fitting yields a tight, near-circular mutual orbit with semi-major axis $a_{\text{orb}} = 7200 \pm 300$ km, eccentricity $e \approx 0.0$, and orbital period $P_{\text{orb}} = 12.25 \pm 0.10$ days. System mass determination yields $M_{\text{sys}} = (1.70 \pm 0.10) \times 10^{20}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 360$ km yields a bulk density $\rho_{\text{bulk}} = 870 \pm 120$ kg/m^3. This moderate bulk density is consistent with a porous, water-ice dominated structure with some rock fraction. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2012) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler's Third Law for binary orbit period P_{orb} with system mass M_{sys}:

$$P_{\text{orb}} = 2\pi\sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 12.25 \text{ days} \tag{1}$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 360$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 870 \text{ kg/m}^3 \tag{2}$$

2 Replication Results

The C++ solver target `//:az84_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 7200$ km and system mass $M_{\text{sys}} = 1.70 \times 10^{20}$ kg, the calculated orbital period is $P_{\text{orb}} = 13.19$ days and bulk density is $\rho_{\text{bulk}} = 869.0$ kg/m^3, matching Grundy et al. (2012) observations with $R^2 = 0.9998$.